PAPER_717: Red Dwarf Compression E: Hydrogen Pages 85-88 and 26-Level Quantum Wave

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Abstract

Doc 43.e (Red Dwarf Compression_E) extends the hydrogen series to pages 85-88, deriving compressed space energy E_{space}, an Earth-Moon tidal analogy using di-pseudo-monopole fields, hydrogen radial probability energies E_{1s} and E_{3d}, and the 26-level quantum wave energy $E_k(t)$ that maps atomic orbital time scales to galactic orbital periods.

1. Compressed Space Energy

$$E_{space} = E_0 \cdot \text{Spatial}_f \cdot \text{Compression} \cdot \text{Layers} \cdot f_{Higgs} \cdot t_{prec} \cdot Q_{scale}$$

with $E_0 = 1.683 \times 10^{-37}$ J (aether base), $f_{Higgs} = 8 \times 10^{-34}$, $t_{prec} = 6.183 \times 10^{-13}$:

$$\boxed{E_{space} \approx 5.52 \times 10^{-104}\,\text{J}}$$

2. Earth-Moon Tidal Analogy

Di-pseudo-monopole (SCm:UA') field analogy to Earth-Moon tidal system:

$$E(t) = E_{aether} \cdot V \cdot \frac{B_{pseudo}^2}{2\mu_0 E_{aether}} \cdot \sin\left(\frac{2\pi t}{T}\right) \cdot f_{spatial}$$

with $B_{pseudo} = 1$ T, $T = 2.36 \times 10^6$ s (orbital period), $f_{spatial} = 2$:

$$E(T/4) \approx 7.96 \times 10^{-22}\,\text{J} \quad \text{(UQFF)}$$

UQFF/SM calibration ratio: $\sim 10^{38}$-10^{39} (SCm:UA di-pseudo-monopole vs SM tidal).

3. Hydrogen Radial Probability

Energies at quarter-period for principal quantum states: | State | $E(T/4)$ | |——-|———-| | $n = 1, l = 0$ (1s) | 3.98×10^{-22} J | | $n = 3, l = 2$ (3d) | 1.99×10^{-22} J |

$$E_{nlm}(t) = E_{aether} \cdot V \cdot \frac{B_{pseudo}^2}{2\mu_0 E_{aether}} \cdot |\psi_{nlm}|^2 r_{max}^2 \cdot \sin\left(\frac{2\pi t}{T}\right)$$

4. 26-Level Quantum Wave

$$E_k(t) = E_{aether} \cdot V \cdot \frac{B_{pseudo}^2}{2\mu_0 E_{aether}} \cdot |Y_{lm}|^2 \cdot \sin\left(\frac{2\pi t}{T_k}\right)$$

where $T_k = \frac{k}{26} \cdot T_{Earth-Moon}$ links atomic timescales to galactic rotation.

Level k	T_k (s)	$E_k(T_k/4)$ (J)
1	9.08×10^4	5.31×10^{-23}
6	5.45×10^5	2.37×10^{-22}

Spherical harmonics used: $|Y_{0,0}|^2 \approx 0.0796$, $|Y_{2,\pm 2}|^2_{max} \approx 0.596$.

References

- UQFF Framework Doc 43.e (Red_Dwarf_Compression_E)
- grok_share_ba508f76c8e.txt entry #66
- Session 176, v5.33

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